

## Highlights

### **Reserve-SOC supervisory energy management for grid-diesel-battery drilling rig power systems under forecast uncertainty**

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- Evidence uses one mechanistic simulator; no target-rig measurements.
- Seven online controllers follow the declared simulation and audit protocol.
- RSS cuts unserved energy 3.27% vs RB and 10.22% vs MLR.
- Composite risk-adjusted cost falls 3.85% vs RB; it is shortage-dominated.
- CR is an independent point-forecast MILP, not a component ablation.

# Reserve-SOC supervisory energy management for grid-diesel-battery drilling rig power systems under forecast uncertainty

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## Abstract

We evaluate reserve-SOC supervisory model predictive control (RSS) for grid-diesel-battery coordination during abrupt load transitions. Evaluation uses a 72 h record and ten independently generated holdouts from a single mechanistic drilling-load and power-system simulator; no synchronized measurements from the target rig are used. Seven online controllers follow a declared comparison protocol; the main contrasts use rule-based control (RB), deterministic margin-loaded reference MPC (MLR), and an independently implemented causal full-MILP (CR). RSS records 813.43 kWh of unserved energy, compared with 840.96 kWh for RB, 906.04 kWh for MLR, and 1,165.81 kWh for CR. These reductions are 3.27%, 10.22%, and 30.23%, with Holm-adjusted exact sign-flip  $p$ -values of 0.02, 0.01, and 0.01. Risk-adjusted cost, a shortage-dominated composite rather than independent economic evidence, falls by 3.85% relative to RB and 9.34% relative to MLR. CR uses one point forecast without the forecast-error envelope or reserve-SOC supervisor. Because CR is weaker than RB under this protocol, the 30.23% RSS-CR gap is interpreted as a bundled boundary comparison, not an isolated component effect. Component interventions associate SOC supervision with a 2.86–4.11 kWh mean per-holdout reduction and the forecast-envelope bundle with a

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further 2.12 kWh reduction. An auxiliary raw evidential risk interface improves selected classification metrics but worsens the Brier score and does not change unserved energy. These findings are limited to the simulated system and do not establish performance for the target rig or drilling rigs generally.

*Keywords:* drilling rig power system, multi-source energy management, battery energy storage, model predictive control, unit commitment, supply reliability

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## 1. Introduction

Drilling rigs are energy-intensive industrial systems whose electrical demand changes sharply with operating state. Drilling, circulation, connection, tripping, and auxiliary activities impose different demand levels and different transition dynamics. When a rig is supplied by a weak grid connection, diesel generator units, and a battery energy storage system (BESS), the operating problem involves more than reducing energy cost. The controller must also maintain power balance, respect generator commitment and ramping constraints, preserve sufficient storage state of charge (SOC), and protect high-priority loads when available supply becomes insufficient.

This problem has a direct operational consequence. A controller that reacts only to instantaneous load may discharge the BESS before a transition, or may start too many diesel units and replace relatively inexpensive grid power with fuel consumption. Conversely, a controller that focuses only on economic dispatch may leave insufficient reserve for a short high-power event. The resulting trade-off is particularly important for drilling rigs because their load profile combines sustained operation with short, high-amplitude transitions.

Energy management studies have investigated hierarchical scheduling, multi-objective microgrid dispatch, energy storage coordination, and model predictive control (MPC). For example, bilevel energy management has been used to coordinate security, reliability, storage, and demand-side management [1], while predictive energy management has been applied to hybrid vessel grids [2]. Stochastic and random-setting formulations have also been used for isolated hybrid microgrids with diesel generation and storage [3]. These studies establish the value of combining forecasting, uncertainty representation, and constrained optimization. They also motivate a system-level evaluation that reports drilling-load transitions, discrete multi-generator

commitment, storage protection, and emergency load prioritization under one declared protocol.

The closest drilling rig precedent considered rule-based generator scheduling together with battery storage for diesel power-plant fuel-efficiency improvement [4]. This prior work motivates the present application, but it also sets a clear comparison requirement: the contribution cannot be reduced to using a BESS in a drilling rig. The relevant question is whether forecast uncertainty and supply risk can be transferred into a controller that remains physically feasible while avoiding unnecessary generator commitment.

This paper develops a reserve-SOC supervisory energy-management framework for a grid-diesel-battery drilling rig power system. The framework links short-horizon load forecasting to scenario-based conditional value-at-risk (CVaR) dispatch, discrete generator commitment, storage reserve management, and priority-based load protection. It addresses three questions: whether the integrated controller changes supply reliability and the associated composite cost, which prespecified component interventions are associated with changes in physical dispatch, and whether the resulting schedules satisfy the stated physical and computational constraints. The paper makes four contributions:

1. We define and audit a causal information boundary that admits only current measurements, executed equipment history, and forecasts available at the decision time, while excluding future true load and future-derived state flags.
2. We use declared component-level interventions to estimate intervention-specific changes associated with the risk input, the reserve-SOC supervisor, and the implemented forecast-envelope bundle under their declared fixed comparison settings.
3. We implement an independent causal rolling full-MILP comparator that shares the realized plant state and cost terms but uses a single point forecast and omits both the forecast-error envelope and reserve-SOC supervisor; it is therefore a bundled boundary comparator rather than a component ablation. Under this protocol, CR has higher unserved energy than RB; this difference cannot be attributed to any single component.
4. We evaluate seven online controllers over ten independently generated simulation holdouts, using seed-level statistics and physical feasibility checks, while reserving the main inferential contrasts for RB, MLR,

and CR. The 72 h record and all holdouts are generated by a single mechanistic simulator; no synchronized target-rig measurements are used, and the conclusions are limited to this simulated system.

The evaluation comprises a 72 h load replay for forecasting and ten independently generated holdouts for the primary dispatch comparison. Separate holdout groups are used for the auxiliary risk interface. Section 3 describes the data provenance and the scope of the resulting conclusions.

## 2. Related work

### *2.1. Drilling rig and industrial hybrid power systems*

Energy storage has been studied in drilling rig power systems as a means of reducing diesel generator inefficiency and smoothing fast load changes. Pavković et al. combined rule-based generator scheduling with a BESS for an oil-drilling diesel power plant [4]. A separate review of drilling rig storage systems discusses the operating and sizing role of storage in drilling applications [5]. For offshore drilling, Huang et al. formulated generation scheduling with uncertain renewable generation and uncertain platform loads [6]. These studies establish the application relevance of coordinated generation and storage. Adjacent offshore studies also assess diesel-electric dispatch with BESS integration using field data [7] and integrated management of offshore oil-and-gas systems with renewable generation and gas turbines [8]. Building on this foundation, the present work connects causal short-horizon forecasting, finite-ensemble shortage risk, discrete multi-unit commitment, SOC supervision, and priority-based protection in one reproducible controller evaluation.

### *2.2. Microgrid energy management and supervisory control*

The broader microgrid literature provides the conceptual foundation for hierarchical supervisory control and energy management. Reviews have classified storage control, microgrid supervisory controllers, and the optimization methods used by energy management systems [9, 10, 11]. A recent review further organizes energy-management strategies by their treatment of forecasting, uncertainty, optimization, and control objectives [12]. Application-specific studies include bilevel security and reliability coordination [1] and random setting management of isolated hybrid microgrids [3]. More recent work covers BESS and renewable-resource coordination [13], affordability–flexibility–security trade-offs [14], MPC-based EMS design [15, 16], two-level

economic MPC [17], and a two-layer robust MPC that exchanges prediction-error-interval information between control time scales [18]. This body of work supports the use of a layered architecture, but the reported results are often organized around a single microgrid formulation or controller. Our distinction is the explicit audit boundary between forecasting, dispatch, SOC mode selection, and emergency load prioritization, together with a declared causal information set and component-level interventions.

### *2.3. MPC, MILP, and risk-aware scheduling*

MPC is well suited to storage-constrained operation because it repeatedly updates decisions as state measurements and forecasts change; a dedicated review surveys MPC architectures and implementation issues for microgrids [19]. Mixed-integer predictive scheduling has been used to account for storage impacts and forecast uncertainty [20], while distributed predictive controllers have coordinated multiple microgrids and hybrid storage systems [21, 22]. Resilience-oriented MPC also makes reserve and supply continuity explicit in the scheduling problem [23]. In parallel, stochastic MILP has been used for hybrid-microgrid energy management [24], and robust cost-risk scheduling has incorporated worst-case CVaR [25]. The CVaR construction used here follows the standard auxiliary-threshold formulation of Rockafellar and Uryasev [26]. Recent studies further report economic and multi-objective MPC [27], systematic MPC-EMS design [28], and BESS-integrated predictive control with experimental evaluation [29]. Scenario-based stochastic MPC [30], risk-aware stochastic control [31], and distributionally robust CVaR scheduling [32] show complementary ways to represent uncertainty. Our contribution is the application-specific reserve-SOC coupling of these tools, supported by component interventions and an independently implemented bundled causal full-MILP boundary comparator.

### *2.4. Evidence fusion for uncertain decisions*

Belief-function methods represent source conflict and residual ignorance in addition to class support. Yager’s rule transfers conjunctive conflict to the full frame of discernment [33], while pignistic transformation supports decisions without treating epistemic imprecision as an ordinary probability [34]. Power-system applications have used evidence fusion for transformer and post-fault network diagnosis [35, 36], and recent work shows that basic-probability-assignment design materially affects diagnostic performance [37].

We use evidence fusion only as an auxiliary risk-reporting interface and evaluate its classification and dispatch effects separately from the primary controller.

### *2.5. Resilience and priority-based critical-load protection*

Priority-aware management is a related but distinct decision layer: it allocates a constrained supply to loads with different service importance, rather than establishing that the total supply shortage has been removed. Recent work combines demand and generation prediction with critical-load management [38], formulates real-time optimization for priority-weighted load restoration [39], and studies resilient scheduling under uncertain demand and storage resources [40]. Hybrid switched MPC with demand-side management also illustrates why supervisory mode selection and controllable demand must be stated separately [41]. Accordingly, the present paper reports priority protection as a diagnostic allocation result and does not infer improved total supply adequacy from the protection-mode flag.

### *2.6. Short-horizon load forecasting*

The forecasting comparison spans naive, tree-based, recurrent, convolutional, and transformer-based candidates. Temporal Fusion Transformers provide an interpretable multi-horizon forecasting design [42]; PatchTST uses patch-based transformer representations for long-term time-series forecasting [43]; and iTransformer reorients the transformer representation for multivariate forecasting [44]. A recent review identifies deep learning choices, online robustness, and feasibility as central concerns for short-term load forecasting [45]. The broader transformer lineage includes efficient sparse attention [46], decomposition and autocorrelation [47], frequency-enhanced decomposition [48], and general temporal-variation modeling [49]. These methods provide strong forecasting baselines. We report their prediction performance separately and then evaluate how forecast errors propagate into generator commitment and storage decisions.

### *2.7. Position of the present study*

The individual ingredients of the proposed framework are established in prior work. The systems question addressed here is how forecast uncertainty, generator commitment, storage reserve, and load priority should interact during rapid drilling-load transitions. We address this question under a common plant, realized-trace, cost, solver, and audit boundary; controller-specific

forecast representations, scenario constructions, and terminal-SOC rules are declared explicitly. Ten primary holdouts support the controller comparison; prespecified interventions estimate local changes associated with the forecast- envelope bundle and SOC supervisor; and an independently implemented causal full-MILP provides a bundled causal optimization boundary reference. The auxiliary evidence-fusion study is reported separately because recognition quality does not necessarily translate into a change in physical dispatch.

### 3. System boundary and data

#### 3.1. System boundary

The system contains four functional components: (i) an external grid connection with time-varying available capacity, (ii) a set of diesel generator units, (iii) a BESS with power and SOC limits, and (iv) an electrical load representing the drilling rig and auxiliary equipment. The controller receives measurements and causally available state information. It produces a receding-horizon plan and applies the first control action before the next measurement update.

The load is partitioned into critical, essential, and interruptible classes for evaluating priority-based protection when available supply is insufficient. The class proportions are control-study settings; deployment values must be approved through the drilling rig’s electrical and safety procedures.

The implementation uses a 5 s sampling interval for the main high-frequency data path. The forecast task uses a 10 min history to predict the following 1 min horizon. The dispatch controller operates on the forecast package and current equipment state; all future true load and future-derived operation-state flags are excluded from online features.

#### 3.2. Evaluation data

The evaluation dataset contains a 72 h active-power record sampled at 5 s, with 51,840 timestamped rows and 51,827 rows after quality processing. It is generated by a mechanistic drilling-load and multi-source power-system simulation parameterized from public process and equipment evidence. The dataset contains total load, operating state, available supply, generator status, battery power, and SOC.

The load generator is a finite-state semi-Markov simulation. If  $Z_j$  is the operating state of segment  $j$ , then its duration is drawn as

$D_j \sim \mathcal{U}(a_{Z_j}, b_{Z_j})$  and the next state is drawn from the fixed categorical transition row  $T_{Z_j, \cdot}$ . The duration ranges in minutes are idle 12–45, circulation 25–100, drilling 20–70, connection 5–12, tripping 35–180, and maintenance 20–90. The nonzero transition probabilities are: idle to circulation/drilling/tripping/maintenance, 0.45/0.10/0.25/0.20; circulation to drilling/idle/tripping/maintenance, 0.65/0.15/0.10/0.10; drilling to connection/circulation/tripping/idle, 0.62/0.23/0.07/0.08; connection to drilling/circulation/tripping/idle, 0.82/0.12/0.02/0.04; tripping to circulation/idle/maintenance, 0.50/0.30/0.20; and maintenance to idle/circulation, 0.70/0.30.

Within a segment, physical load is

$$P_t = P_t^{aux} + P_t^{mud} + P_t^{top} + P_t^{draw} + P_t^{other}, \quad (1)$$

where each component combines a state-specific rated-power fraction, a periodic or triangular operating term, and a Gaussian innovation; drilling terms are additionally modulated by simulated depth and formation hardness. All components within each state are rescaled by one frozen factor to the public-evidence target mean and then constrained by the declared state cap. The recorded power channel is

$$Y_t = Q_{0.5}(\text{clip}\{P_t(1 + r_t), 180, 2000\}), \quad r_t = \text{clip}\{w_t + b_t, -0.025, 0.025\}, \quad (2)$$

where  $w_t$  is Gaussian white relative error with standard deviation 0.005, and  $b_t$  is a Gaussian-innovation AR(1) relative drift with coefficient 0.9995 and standard deviation 0.003. A daily Bernoulli trial with probability 0.35 introduces a missing block whose start is uniform within the day and whose duration is discrete uniform from 15 to 90 s; these short gaps are linearly interpolated.

The frozen calibration map records a source identifier and provenance class for each state-load target, equipment quantity, and cost field. Public drilling generator and storage observations anchor the fleet and BESS ranges [4, 5]; state means and caps are mapped to the corresponding public load sources in the calibration file. Transition probabilities, duration ranges, generator ramp and dwell settings, startup and degradation costs, and the missingness model are marked as engineering scenario assumptions rather than distributions fitted to the target rig. Exact state-specific coefficients and transformations are fixed in the versioned generator and resolved config-

uration retained with the reproducibility artifacts. Calibration reads neither forecasts nor controller outcomes.

The forecasting replay contains 51,709 samples, of which 51,697 are scored. A fixed chronological split is used for model comparison. The power scaler is fitted on the training partition, ensemble weights and forecast-error settings use the validation partition, and the test targets are excluded from model fitting and selection.

For dispatch, each forecast origin retains one 12-step prediction row for each model. The seven-member scenario package consists of Persistence, Validation-Weighted-Ensemble (VWE), PatchTST, iTransformer, StateAware-TCN-Attention, StateAware-Patch-Transformer, and StateAware-DualBranch-Patch-Transformer. Persistence repeats the last observed load over all 12 steps. VWE is a nonnegative convex combination of XGBoost, LSTM, TCN, and the three state-aware models; its weights sum to one and are fitted on validation data by minimizing mean squared error plus an  $L_2$  penalty of  $10^{-4}$ . The other five members enter the package as their direct multi-step outputs. Test targets are not used to construct or weight these members.

An official public Wellsite Information Transfer Standard Markup Language (WITSML) example provides the drilling process and state-signal schema used in the interface design. Because the example does not contain synchronized total active power, it is not included in the controller-performance evaluation.

Figures 1 and 2 summarize the load features and representative operating-state curves used to define the forecasting and dispatch problem; Table 1 summarizes the evaluation datasets and their roles.

### 3.3. Preprocessing and causality

The preprocessing path canonicalizes timestamps, checks the sampling interval, records quality flags, and imputes only the documented short gaps. The forecast input window is constructed from information available at the prediction timestamp. Transition features are based on history, not on the future operation-state label. This distinction is essential because a future-derived transition flag would make the forecast or controller appear stronger while being unavailable during online operation.

Development cases and holdout cases are separated throughout the evaluation. Controller settings and eligibility criteria are fixed before holdout

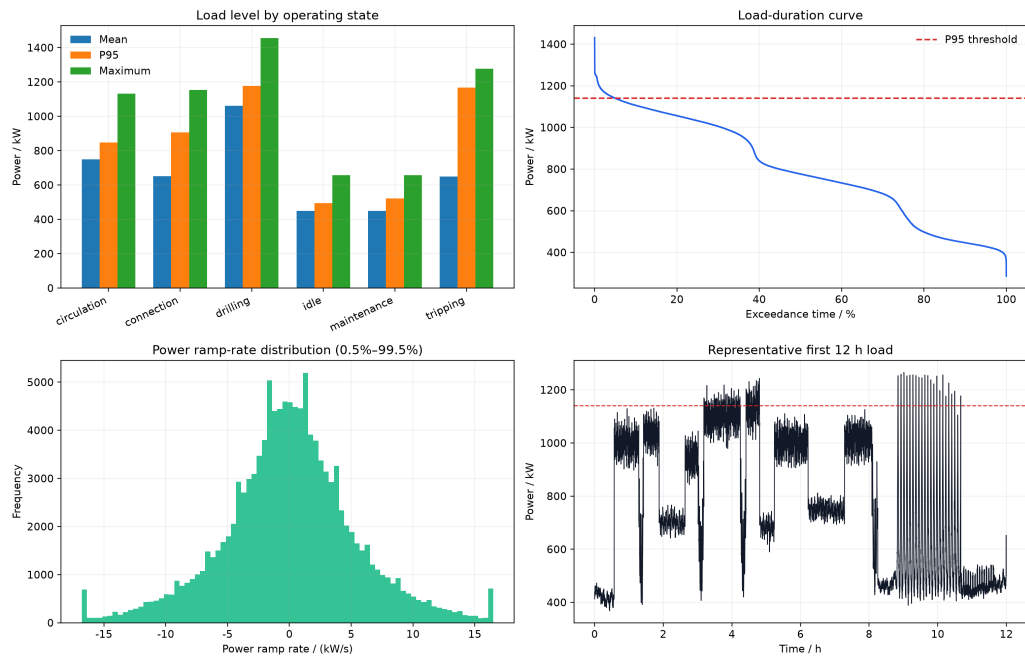


Figure 1: Load-feature summary across operating states and representative intervals

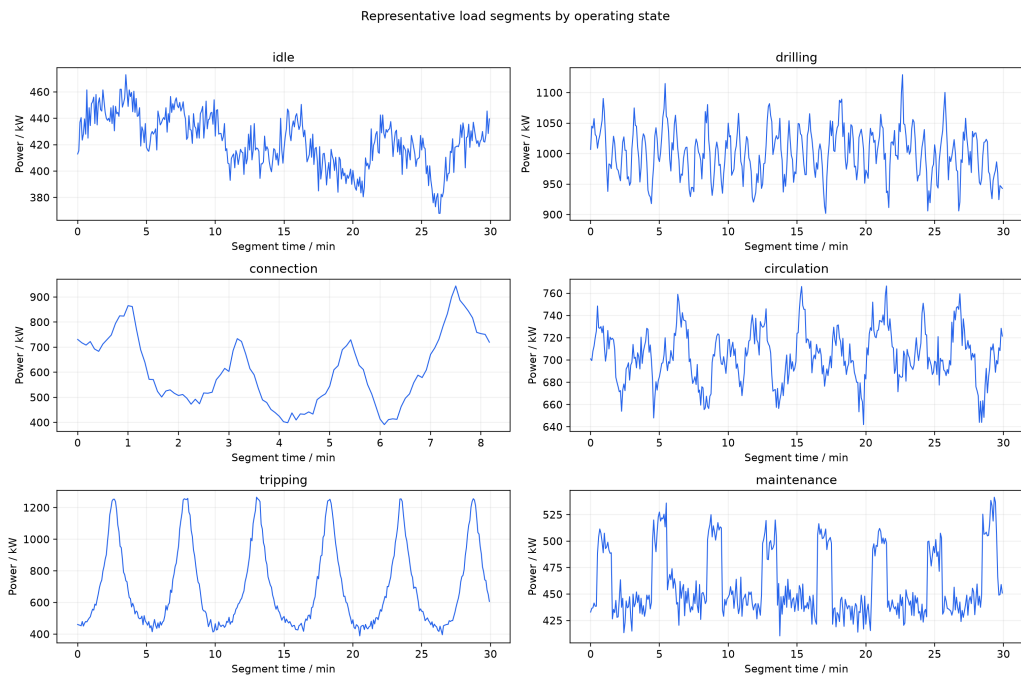


Figure 2: Representative load curves for six drilling operating states

Table 1: Evaluation datasets and their roles

| Dataset / source        | Role                                | Resolution       | Content and provenance                                                                   |
|-------------------------|-------------------------------------|------------------|------------------------------------------------------------------------------------------|
| Rig power-system record | Forecasting and dispatch evaluation | 5 s              | Load, supply, generator, storage, operating-state, and risk variables                    |
| Forecasting replay      | Chronological model comparison      | 5 s              | Fixed train, validation, and test partitions from the evaluation record                  |
| Public WITSML example   | Process-signal interface design     | source-dependent | Public drilling process and operating-state fields; no synchronized active power channel |

outcomes are inspected, and no parameter is retuned during the ten-case primary comparison.

For each holdout, its seed initializes the state sequence and durations, state-specific amplitudes and Gaussian innovations, depth/formation evolution, transient timing and amplitude, and missing-block process. A deterministic derived seed, equal to the holdout seed plus 14,000,003, initializes the post-calibration measurement error. Distinct holdout seeds therefore define separate pseudorandom realizations of the same frozen simulator; they are not claimed to be draws from an empirically fitted target-rig population.

All seven member rows at a dispatch decision share the same causal forecast origin and horizon. Source indices are required to be consecutive before a dispatch window is selected. For CEF, a test-period residual can update the error-feedback state only after its complete 12-step target window has been observed; at forecast origin  $i$ , the newest eligible residual is therefore from origin  $i - 12$ . For each holdout, the simulator random-number generator is initialized with the holdout seed, and the forecasting benchmark initializes NumPy and PyTorch once with the same seed before fitting models in the fixed implementation order; XGBoost receives that seed as its random state. The seven-member stacking and subsequent dispatch introduce no random draw and are not reseeded by scenario or decision time.

## 4. Reserve-SOC supervisory coordination framework

### 4.1. Forecasting module

Let  $L_t$  denote the total active load at time  $t$ , and let  $H$  be the forecast horizon. The forecasting module produces a point forecast in Equation (3).

$$\hat{\mathbf{L}}_{t+1:t+H} = f_{\theta}(\mathbf{X}_{t-W+1:t}), \quad (3)$$

where  $W$  is the historical input length and  $\mathbf{X}$  contains the available load, equipment, and causal process features. The comparison includes Persistence, XGBoost, long short-term memory (LSTM), temporal convolutional network (TCN), PatchTST, iTransformer, and state-aware ensemble variants. The causal error-feedback ensemble supplies the point forecast used by the controller, while iTransformer provides the strongest forecasting baseline in the replay [42, 43, 44].

For control, the point forecast is combined with a forecast-error envelope. In generic form, Equation (4) applies.

$$\mathbf{L}_{t+1:t+H}^{(s)} = \hat{\mathbf{L}}_{t+1:t+H} + \boldsymbol{\epsilon}_{t+1:t+H}^{(s)}, \quad (4)$$

where  $s$  indexes members and  $\boldsymbol{\epsilon}^{(s)}$  is the deterministic difference between member  $s$  and the CEF point forecast. It is not a sampled error from a fitted probability distribution. Scenario construction is performed without using future true values as online control features.

### 4.2. Supply, storage, and load variables

At each control step, the decision variables include grid power  $P_t^g$ , generator power  $P_{i,t}^{gen}$ , unit commitment  $u_{i,t}$ , storage charging power  $P_t^{ch}$ , storage discharging power  $P_t^{dis}$ , and nonnegative unserved-load variables  $U_t$ . The BESS SOC is  $S_t$ , and  $\eta_{ch}$  and  $\eta_{dis}$  denote charging and discharging efficiencies. For a priority class  $k$ ,  $U_{k,t}$  is the load in that class that is not served, so  $0 \leq U_{k,t} \leq L_{k,t}$  and  $U_t = \sum_k U_{k,t}$ .

The physical balance for a scenario is represented by Equation (5).

$$L_t^{(s)} - U_t^{(s)} = P_t^{g,(s)} + \sum_i P_{i,t}^{gen,(s)} + P_t^{dis,(s)} - P_t^{ch,(s)}, \quad (5)$$

where  $U_t^{(s)}$  is nonnegative unserved load. This convention makes the served load equal to  $L_t^{(s)} - U_t^{(s)}$  and is also used when computing unserved energy. The storage state is updated as shown in Equation (6).

$$S_{t+1} = S_t + \frac{\eta_{ch} P_t^{ch} \Delta t}{E_{BESS}} - \frac{P_t^{dis} \Delta t}{\eta_{dis} E_{BESS}}, \quad (6)$$

with SOC bounds, charge/discharge exclusivity, and unit-commitment constraints. The dispatch configuration uses a 5 s control interval, a 30 min evaluation window with a 10 min warm-up, a 1300 kW grid limit, four 240 kW generator units, a 500 kW/133 kWh BESS, initial SOC 0.65, SOC bounds [0.15, 0.90], and charge/discharge efficiencies of 0.95. The unit minimum stable output is 120 kW, the unit ramp limit is 30 kW per control step, and the minimum up/down times are 12/6 steps. The cost configuration uses 0.71 yuan/kWh for grid energy, 7.50 yuan/L for diesel, 0.15 yuan/kWh for storage degradation, 100 yuan/kWh for unserved energy, and 5 yuan per generator startup. These values define the evaluation case and should be replaced with site measurements during deployment. The optimizer uses a linear fuel-cost approximation; reported fuel consumption is recomputed with the declared piecewise generator fuel curve. Dispatch is solved with HiGHS 1.12.0 through SciPy 1.17.1, using a 5 s time limit and a  $1.00 \times 10^{-6}$  relative MIP gap. The recorded decision latency covers forecast-to-dispatch computation.

For each generator unit, the MILP enforces the minimum-stable and rated-power bounds, the ramp limits, and the logical relation between commitment, startup, and shutdown variables. Minimum up/down constraints include the executed transition history at the beginning of each horizon. The first control action is shared across forecast scenarios, whereas later continuous recourse variables may differ by scenario. This prevents the controller from using future scenario identity to choose the first physical action.

#### 4.3. Dispatch objective

The receding-horizon controller minimizes expected operating cost plus a CVaR penalty on shortage cost over the current horizon in Equation (7):

$$\begin{aligned} \min J = \mathbb{E}_s \Big[ & J_{energy}^{(s)} + J_{fuel}^{(s)} + J_{start}^{(s)} + J_{deg}^{(s)} \\ & + J_{short}^{(s)} + J_{spill}^{(s)} \Big] + \beta \text{CVaR}_\alpha \left( J_{short}^{(s)} \right). \end{aligned} \quad (7)$$

Here  $J_{short}^{(s)}$  is the shortage penalty for scenario  $s$ ,  $J_{spill}^{(s)}$  penalizes unused surplus, and the scenario probability is  $1/S$ . The CVaR term is implemented with an auxiliary threshold  $\eta$  and excess variables  $\xi_s$  in Equation (8):

$$\text{CVaR}_\alpha(J_{short}) = \eta + \frac{1}{(1-\alpha)S} \sum_{s=1}^S \xi_s, \quad \xi_s \geq J_{short}^{(s)} - \eta, \quad \xi_s \geq 0. \quad (8)$$

This auxiliary-variable representation is the standard linearizable CVaR construction [26]; the stochastic-MILP and robust cost-risk literature provides related scheduling precedents [24, 25].

The scenario set contains seven forecast members, uses  $\alpha = 0.90$  and  $\beta = 1.00$  at the base risk level, and changes these values by risk level to  $(0.90, 1.00)$ ,  $(0.90, 1.00)$ ,  $(0.93, 1.50)$ , and  $(0.95, 2.00)$ . Because the seven members have equal weight and every reported  $\alpha$  exceeds  $1 - 1/7$ , the empirical CVaR term in Equation (8) equals the largest member shortage cost. The reported changes in  $\alpha$  therefore do not select different empirical tails. Within the CVaR term, risk-level scaling operates through  $\beta$ ; risk level also changes the controller-specific terminal-SOC rule reported in Section 5.4. Accordingly, this term is interpreted as a finite-ensemble worst-member shortage penalty, not as a calibrated probabilistic coverage statement. The commitment variables are integer and include generator availability, minimum stable output, ramping, minimum up/down time, and startup/shutdown logic. The seven online-controller comparisons share the plant model, realized load and capacity traces, price terms, execution layer, and physical audit path. Forecast representation, scenario construction, and terminal-SOC rules are controller-specific and are declared in Section 5.4. Risk-adjusted cost is computed from the resolved cost configuration after the same physical replay; it is not inferred from fuel alone.

#### 4.4. Reserve-SOC supervisory layer

The supervisory layer selects the operating mode around the rolling dispatch controller while retaining the constraints of the physical model. It uses current SOC, available capacity, the causal risk signal, and the forecasted supply-demand envelope. In normal operation, the controller uses risk-adaptive CVaR MPC; under protection, it uses robust MPC. Protection begins when the risk level is at least 2 or SOC is at most 0.30. Normal operation resumes when the risk level is at most 1, SOC is at least 0.35, and both conditions persist for 12 control steps.

When protection is active and SOC is at most 0.30, the controller requests at least two reachable generator units if the forecast-deficit guard is triggered.

A grid-headroom rule targets SOC 0.75 and limits the storage adjustment to 20% of rated storage power. This rule preserves reserve without starting an additional generator solely to charge the battery.

#### *4.5. Auxiliary risk interface*

The auxiliary evidential variant (E-RSS) changes only the discrete risk signal supplied to RSS. It combines learned risk scores, forecast margin and model disagreement, current SOC, grid derating, and the causal forecast ramp over four states: normal, watch, warning, and severe. Source reliability is applied by Shafer discounting, Yager combination retains conflict as ignorance, and a pignistic score supplies the discrete controller input. Conflict and ignorance remain diagnostic outputs. The forecast, plant, dispatch model, SOC supervisor, prices, and scenario settings are identical to RSS.

A follow-up diagnostic compares Yager combination with normalized Dempster combination and temperature calibration. Raw Dempster scores drive the D-RSS dispatch variant; scalar calibration with  $T = 0.50$  produces DC-RSS probability reports without changing the selected class. This separation allows probability calibration and classification quality, physical actions, and energy-system outcomes to be evaluated independently. Section 6.6 reports this diagnostic because the evidential interface is not part of the primary reliability claim.

#### *4.6. Emergency load protection*

When total available supply cannot satisfy all loads, the emergency module allocates supply by priority. Critical loads are protected first, necessary loads are protected second, and interruptible loads absorb the remaining deficit. The controller reports both critical load unserved energy and total unserved energy. A reduction in critical load shortage does not imply that total shortage has been eliminated.

Table 2 summarizes the symbols, constraints, and parameter provenance used by the dispatch formulation.

## **5. Experimental protocol**

### *5.1. Research questions*

The experiments address three questions:

**RQ1.** How do the forecasting candidates compare on total-load, peak, and transition errors under the same replay?

Table 2: Symbols and constraints

| Symbol / group        | Meaning                                                      | Unit      | Parameter provenance / configuration                                                                                    |
|-----------------------|--------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------|
| $L_t$                 | total rig active load                                        | kW        | mechanistic evaluation record                                                                                           |
| $P_t^g$               | grid active power                                            | kW        | 1300 kW grid limit                                                                                                      |
| $P_{i,t}^{gen}$       | generator $i$ active power                                   | kW        | four 240 kW units                                                                                                       |
| $u_{i,t}$             | generator commitment                                         | binary    | MILP formulation                                                                                                        |
| $P_t^{ch}, P_t^{dis}$ | BESS charging and discharging power                          | kW        | 500 kW BESS limit                                                                                                       |
| $S_t$                 | BESS state of charge                                         | % or p.u. | initial 0.65; bounds [0.15, 0.90]                                                                                       |
| $U_t$                 | nonnegative unserved load; sum of class-level curtailed load | kW        | priority-based allocation setting                                                                                       |
| $\Delta t, H, W$      | time step, forecast horizon, history length                  | s/steps   | 5 s; 12-step horizon; 120-step history                                                                                  |
| $\beta, \alpha$       | CVaR weight and empirical-tail parameter                     | —         | Seven equal-weight members; all reported $\alpha$ values select the same worst-member tail; $\beta = 1.00$ at base risk |
| $m_Y, \text{BetP}$    | Yager-combined mass and pignistic decision score             | —         | E-RSS interface; ignorance and conflict reported separately                                                             |
| $K, m_D, p_T$         | conjunctive conflict, Dempster mass, calibrated score        | —         | D-RSS/DC-RSS follow-up; $T = 0.50$ , raw score drives dispatch                                                          |

**RQ2.** Does RSS reduce unserved energy and the associated composite risk-adjusted cost relative to rule-based control, deterministic margin-loaded

reference MPC, and the bundled causal full-MILP boundary comparator?

**RQ3.** Which control components change the physical dispatch, and does the controller remain feasible under generator, grid, storage, missing-data, and telemetry disturbances?

RQ3 is examined through a  $2 \times 2$  risk-by-supervisor intervention, a forecast-envelope intervention, auxiliary risk-interface diagnostics, emergency load-allocation tests, and physical feasibility checks.

## 5.2. Baselines

The forecasting baselines are Persistence, XGBoost, LSTM, TCN, PatchTST, iTransformer, and state-aware or ensemble variants. This set includes a naive reference, classical machine learning, recurrent neural networks, convolutional sequence modeling, and recent transformer-based models. For compact tables, the longer forecasting labels are abbreviated as follows: state-aware patch transformer (SA-PT), causal error-feedback ensemble (CEF), validation-weighted ensemble (VWE), state-aware dual-branch patch transformer (SA-DB), and state-aware TCN-attention (SA-TA).

The dispatch baselines are rule-based control (RB), persistence MPC (P-MPC), deterministic margin-loaded reference MPC (MLR), scenario CVaR MPC (SC), risk-adaptive CVaR MPC (RAC), risk-aware ensemble MPC (RAE), and the proposed reserve-SOC supervisory MPC (RSS). RB captures a transparent operational policy; MLR provides a deterministic single-path conservative-demand reference; the CVaR variants provide the prespecified scenario and adaptive-risk settings; and RSS represents the complete primary chain. The Yager evidential extension is E-RSS; the Dempster and Dempster-calibrated variants are D-RSS and DC-RSS. The independent causal rolling full-MILP and perfect-foresight full-horizon MILP references are denoted CR and PF, respectively.

All online controllers share the plant model, realized load and capacity traces, price terms, execution layer, and physical audit path. Forecast representation, scenario construction, and terminal-SOC rules are controller-specific and are declared in Section 5.4. Two independently implemented MILP references supplement these comparisons: a causal rolling full-MILP that uses the current 12-step point forecast, and a deterministic full-horizon solution with perfect load information. CR provides a bundled causal boundary comparison for evaluating the causal optimization reference under the complete declared information boundary; it is not used for component-level

attribution. The perfect-foresight result provides a best-case operating reference.

### 5.3. Metrics

Forecast metrics include mean absolute error (MAE), root mean squared error (RMSE), the coefficient of determination ( $R^2$ ), normalized MAE, sMAPE, peak MAE, and transition MAE. Transition MAE is reported because average error can hide the short events that determine generator commitment and storage reserve.

Dispatch metrics include total and per-scenario unserved energy, risk-adjusted cost, equivalent energy cost, diesel fuel, generator startups, maximum decision latency, and physical audit failures. Risk-interface metrics include Macro-F1, balanced accuracy, high- and severe-risk recall, multiclass Brier score, mean ignorance, mean conflict, changed risk levels, and changed physical actions. Physical audits check power balance, dynamic grid/generator/storage capacity, SOC bounds, simultaneous charge/discharge, integer commitment, and solver/runtime validity.

The main comparison uses unserved energy as the primary reliability metric and risk-adjusted cost as its shortage-dominated composite monetary summary. Because that cost includes the shortage penalty, it is not treated as an independent economic outcome. Fuel, startups, latency, terminal SOC, and shortage are reported together so that lower fuel use is not obtained by prematurely depleting the battery or leaving load unserved.

### 5.4. Reproducibility protocol

Development and holdout cases are separated before controller comparison. The primary analysis uses ten eligible holdouts and treats the holdout seed as the independent unit, with three operating scenarios clustered within each seed. A method-scenario trajectory denotes one controller run for one holdout seed and one operating-scenario window. The resulting dataset contains 210 main method-scenario trajectories. Exact two-sided sign-flip tests, 20,000-resample seed bootstrap intervals, and Holm correction across the three primary controller contrasts are applied at seed level.

The primary seeds are 20261011, 20261012, and 20261014–20261021. Development seeds 20261008 and 20261009 and diagnostic seed 20261004 are excluded from primary inference. Candidate seed 20261013 failed the frozen public-calibration overall-maximum-power gate before dispatch; no

controller was executed for that seed. The subsequent queue 20261014–20261017 supplied the first two eligible replacements, and the separately frozen queue 20261016–20261027 stopped after its first six eligible seeds, 20261016–20261021; 20261022–20261027 were not run for the primary comparison.

Eligibility is determined before any controller outcome. The public-evidence calibration audit must pass, and the generated trace must contain at least one 360-step window for each declared operating case. Case A requires at least 0.90 drilling rows, 0.60 normal-supply rows, and no emergency-supply row. Case B requires at least 0.75 drilling-or-connection rows, 0.10 connection rows, 0.50 constrained-or-weak-supply rows, and at most 0.20 emergency rows. Case C requires at least 0.90 tripping rows and 0.50 weak-or-emergency-supply rows. Candidate windows are scanned at an 18-step stride. These rules and the ordered stopping rule are frozen independently of controller performance.

*Forecast scenarios..* At forecast origin  $t$ , the implementation forms a  $7 \times 12$  array by stacking the seven stored member rows listed in Section 3.2, clips negative demand to zero on entry to the scenario optimizer, and assigns each member probability  $1/7$ . The members are model-disagreement trajectories, not bootstrap or Monte Carlo perturbations of CEF. Consequently, scenario construction has no external forecast-error sample pool, no fitted parametric distribution, and no residual-resampling seed. In the notation of the forecasting section,  $\epsilon^{(s)}$  is the deterministic difference between member  $s$  and the CEF point forecast, rather than a sampled error. Validation residuals are used separately to select the CEF feedback parameters and to store a horizon-wise upper-envelope artifact; neither operation samples the seven CVaR members. During test replay, CEF releases only fully observed, 12-step-delayed residuals. With seven equal weights, the reported  $\alpha$  values all reduce the empirical CVaR term to the maximum member shortage cost, as stated in Section 4.3; the members are not interpreted as calibrated probability samples.

*Supervisor state machine..* The supervisor is evaluated before each 5 s action using the current predicted risk level and pre-decision SOC. Each scored window starts with protection inactive and the safe-exit counter equal to zero; the shared warm-up transfers plant and commitment state, but not supervisor mode. Entry is immediate when risk is at least 2 or SOC is at

most 0.30, with equality included. Once active, the counter increments only on a row satisfying both risk at most 1 and SOC at least 0.35. Any row that fails either exit condition resets the counter to zero; an entry-condition row also keeps protection active and resets it. Protection is released on the 12th consecutive safe row, and the counter is then reset. Thus an inactive controller remains inactive in the hysteresis band unless an entry threshold is met, whereas an active controller remains active there.

*Terminal SOC.* Every rolling optimization recomputes its terminal lower bound from the pre-decision SOC as  $S_{t+H}^{(s)} \geq \max\{S_{\min}, S_t - \delta\}$ , together with the physical upper bound  $S_{t+H}^{(s)} \leq S_{\max}$ ; the constraint is imposed for every scenario where applicable. RB has no terminal optimization constraint. The fixed drops are  $\delta = 0.025$  for P-MPC and 0.005 for MLR, SC, and CR. For RAC and the economic mode of RSS, the values for risk levels 0–3 are 0.005, 0.010, 0.020, and 0.030, respectively; RSS protection mode uses the MLR value 0.005. For RAE, the corresponding values are 0.002, 0.005, 0.015, and 0.030. This is a rolling depletion limit relative to current SOC, not a fixed end-of-window SOC target.

*Baseline implementations.* RB uses the measured current load. It assigns grid power first, then generator power up to the smaller of the remaining deficit, available generator capacity, and 75% of the 960 kW aggregate rating, and then battery discharge for any remaining deficit. With no grid-side deficit and SOC below 0.75, it charges from unused grid capacity up to the smaller of available battery power and 20% of the 500 kW storage rating. The common execution layer then enforces current capacity, SOC, commitment, ramp, and minimum up/down limits and resolves the realized power balance. P-MPC defines persistence by  $\hat{L}_{t+h|t} = L_t$  for  $h = 1, \dots, 12$ , solves the same rolling clustered unit-commitment/storage model, and applies its first action. The deterministic margin-loaded reference MPC (MLR) uses the single deterministic demand path  $\bar{L}_{t+h|t} = \max\{0, \hat{L}_{t+h|t}^{\text{CEF}} + 10 + 0.25|\hat{L}_{t+h|t}^{\text{CEF}} - \hat{L}_{t+h|t}^{\text{P}}|\}$ . MLR optimizes only this margin-loaded path; it does not construct a set-valued uncertainty region or solve an additional min–max problem. It is therefore a deterministic single-path formulation rather than a set-valued or min–max robust MPC formulation.

*Shared warm-up and CR boundary handling.* The online benchmark requests 120 pre-window steps and truncates the warm-up to the available contiguous prefix when a selected window begins earlier. The realized load and

capacity trace are replayed under RB, and warm-up actions are excluded from scoring. Among the 30 primary windows, 29 use 120 steps and the B connection-transition window for seed 20261019 uses 90. CR does not generate a separate warm-up trajectory: it imports the shared post-warm-up SOC, generator output, committed-unit count, the most recent 11 startup counts, and the most recent five shutdown counts. The export routine checks SOC, generator output, and committed units against the frozen window meta-data before CR is run. CR-specific unit tests cover a recent startup that remains locked on, shutdown when no startup lock exists, forced capacity derating that removes an infeasible minimum-up lock, and netting historical startup/shutdown counts to the currently online units. Supervisor tests separately cover entry at the risk threshold, continued protection through 11 safe rows, exit on row 12, low-SOC entry, and counter reset by an unsafe row.

The two auxiliary risk-interface studies use separate six-holdout groups and do not alter the primary comparison. Their settings, eligibility rules, and stopping criteria are fixed before the corresponding outcomes are inspected. Classification, calibration, physical action, cost, and unserved energy are reported separately. Versioned configurations, ordered seed lists, hashes, and trajectory-level audit results are retained in the reproducibility package.

Table 3 summarizes the experimental comparison protocol.

The forecast models use 120 historical steps (10 min) to predict 12 steps (1 min), batch size 512, a maximum of 20 training epochs, early-stopping patience of 5, learning rate  $8.00 \times 10^{-4}$ , weight decay  $1.00 \times 10^{-4}$ , and seed 20260902. The replay was run with Python 3.11.16, PyTorch 2.9.1, XGBoost 3.2.0, and an NVIDIA GeForce RTX 5060 Laptop GPU.

## 6. Results

### 6.1. Forecasting comparison

Table 4 summarizes the forecasting replay. iTransformer obtains the lowest MAE (56.51 kW), lowest RMSE (115.62 kW), and highest  $R^2$  (0.94) among the listed candidates. Its MAE is 52.55% lower than Persistence. PatchTST and the state-aware transformer are the next strongest candidates by average MAE, while the causal error-feedback ensemble remains competitive but does not surpass iTransformer.

The forecasting comparison serves two purposes. It places modern transformer forecasts in the scenario package used by the dispatch controller, and

Table 3: Experimental comparison protocol

| Item              | Forecast comparison                                                                             | Dispatch comparison                                                                                  |
|-------------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Evaluation basis  | Chronological load replay                                                                       | Power-system simulation                                                                              |
| Sampling          | 5 s                                                                                             | 5 s control/evaluation path                                                                          |
| Forecast window   | 10 min history $\rightarrow$ 1 min horizon                                                      | 12-step horizon; forecast representation is controller-specific                                      |
| Baselines         | 11 candidates including iTransformer                                                            | 7 online controllers plus CR; E-RSS, D-RSS, and DC-RSS are auxiliary variants; PF is diagnostic only |
| Main split        | chronological replay: 25,920 train / 8,640 validation / 17,280 test rows; 51,697 scored samples | ten primary holdouts; three scenarios clustered per seed                                             |
| Primary metrics   | MAE, RMSE, $R^2$ , transition MAE                                                               | unserved energy, risk-adjusted cost                                                                  |
| Additional checks | peak error, runtime                                                                             | physical audit, SOC, latency, per-scenario regret                                                    |
| Reproducibility   | fixed split and software environment                                                            | fixed controller settings, seed-level statistics, and physical audits                                |

it quantifies peak and transition errors that affect generator commitment and storage reserve. CR uses CEF alone to maintain a transparent point-forecast information boundary. The strongest forecast is an existing iTransformer baseline; the contribution evaluated below is the propagation of forecast uncertainty into the energy-management decision.

Figure 3 separates overall behavior from the principal target transition and shows only four prespecified representative predictors: iTransformer (best mean error), SA-PT (best transition MAE), CEF (the controller’s causal point forecast), and Persistence (the naive reference). Figure 4 reports all forecasting candidates using the abbreviations defined in Section 5.2.

### 6.2. Ten-holdout dispatch comparison

Table 5 reports aggregate results for all seven online controllers across ten independently generated eligible holdouts. The prespecified primary inferential contrasts in this subsection are RSS versus RB and MLR. SC and RAC also serve as factorial arms in the mechanism analysis, whereas P-MPC

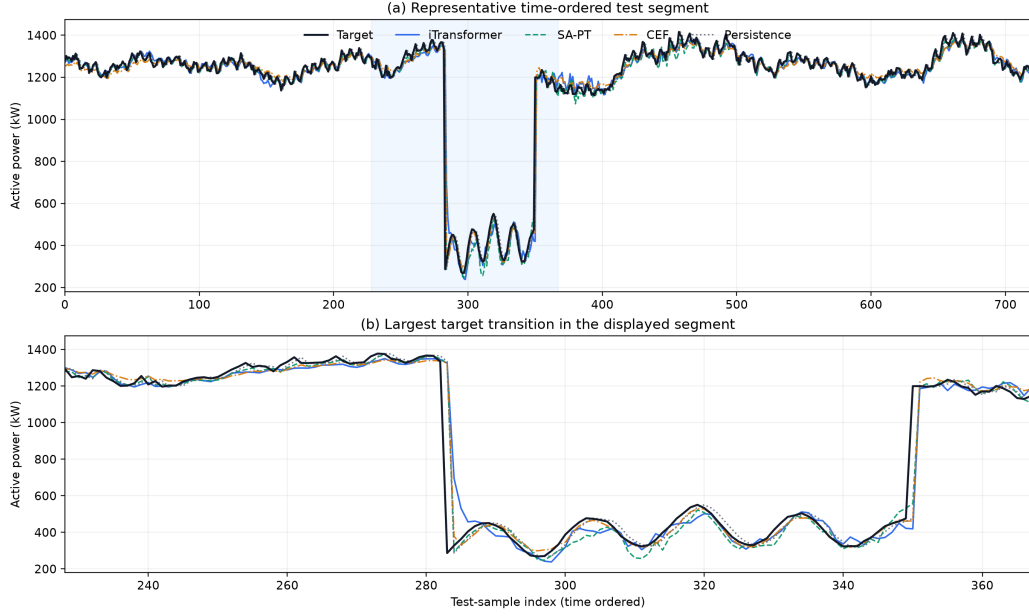


Figure 3: Representative forecast traces and transition detail

Table 4: Forecasting comparison on the evaluation replay

| Model        | MAE (kW) | RMSE (kW) | $R^2$ | Peak MAE (kW) | Transition MAE (kW) |
|--------------|----------|-----------|-------|---------------|---------------------|
| iTransformer | 56.51    | 115.62    | 0.94  | 46.42         | 361.82              |
| PatchTST     | 59.47    | 118.30    | 0.94  | 55.85         | 388.97              |
| SA-PT        | 60.96    | 120.84    | 0.93  | 58.76         | 359.88              |
| CEF          | 61.89    | 120.22    | 0.93  | 63.70         | 394.73              |
| TCN          | 66.43    | 125.62    | 0.93  | 51.49         | 377.38              |
| VWE          | 68.27    | 126.15    | 0.93  | 57.85         | 388.13              |
| SA-DB        | 72.41    | 129.55    | 0.92  | 68.87         | 370.53              |
| SA-TA        | 74.75    | 144.99    | 0.90  | 63.51         | 370.95              |
| XGBoost      | 87.09    | 156.91    | 0.89  | 63.40         | 371.29              |
| LSTM         | 98.01    | 162.78    | 0.88  | 88.28         | 450.37              |
| Persistence  | 119.10   | 238.76    | 0.74  | 66.13         | 460.86              |

and RAE are retained as secondary descriptive comparators. RSS produces 813.43 kWh of unserved energy, compared with 840.96 kWh for RB and 906.04 kWh for MLR, corresponding to reductions of 3.27% and 10.22%, respectively. Its aggregate risk-adjusted cost is 95,366.92 yuan, 3.85% below RB and 9.34% below MLR.

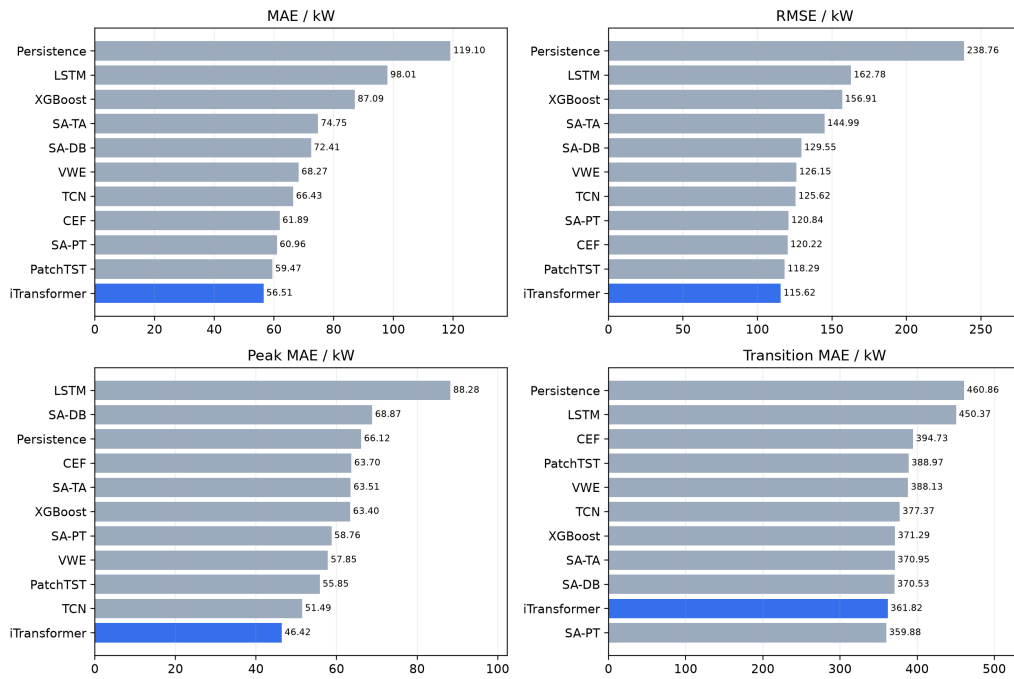


Figure 4: Forecast metric comparison for all candidates

Risk-adjusted cost is not independent economic evidence. It includes the declared 100 yuan/kWh unserved-energy penalty, which contributes approximately 85% of the reported RSS, RB, and MLR totals. Consequently, the cost comparison is mechanically coupled to the reliability comparison and is interpreted as a composite monetary summary rather than a second independent outcome. RSS uses 37.20 L less diesel than RB but 2.61 L more than MLR; fuel use is therefore reported separately from the composite cost.

The seed is the independent analysis unit; the three scenarios remain clustered within each seed. RSS minus RB is  $-2.75 \pm 5.39$  kWh (mean  $\pm$  sample SD; seed-bootstrap 95% interval,  $-6.36$  to  $-0.29$  kWh), and RSS minus MLR is  $-9.26 \pm 11.31$  kWh (interval,  $-16.31$  to  $-3.33$  kWh). RSS is lower or tied on every seed: 7 improvements and 3 ties against RB, and 9 improvements and 1 tie against MLR. Exact two-sided sign-flip  $p$ -values are 0.02 and  $< 0.01$ ; the corresponding Holm-adjusted values are 0.02 and 0.01. All 210 main online method–scenario trajectories pass the physical audit, and the maximum RSS decision time is 0.36 s.

Figure 5 shows an illustrative dispatch and SOC trajectory during a connection-transition replay. To retain legibility at manuscript size, it displays the three prespecified main-comparison methods (RB, MLR, and RSS). Table 5 reports all seven online controllers; SC and RAC are revisited as mechanism arms, while P-MPC and RAE remain descriptive comparators.

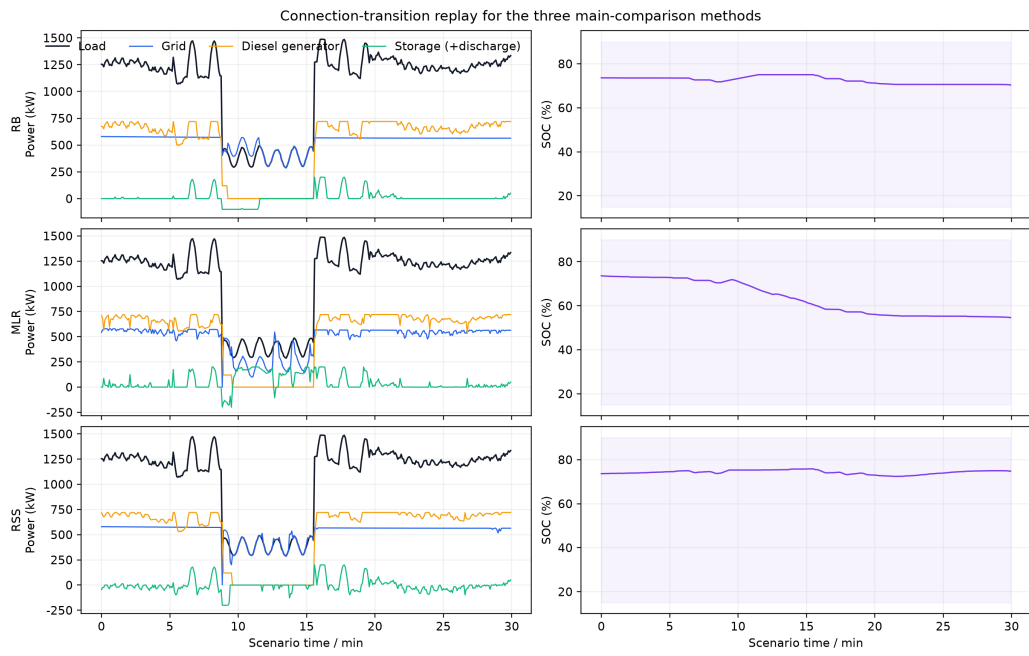


Figure 5: Dispatch and SOC behavior of RB, MLR, and RSS during a connection transition

Table 5: Aggregate ten-holdout dispatch results for all seven online controllers

| Method | Unserved energy (kWh) | Risk-adjusted cost (yuan) | Diesel fuel (L) | Maximum decision time (s) |
|--------|-----------------------|---------------------------|-----------------|---------------------------|
| RSS    | 813.43                | 95,366.92                 | 722.32          | 0.36                      |
| RB     | 840.96                | 99,189.91                 | 759.52          | 0.00                      |
| MLR    | 906.04                | 105,192.33                | 719.71          | 0.12                      |
| P-MPC  | 1,070.88              | 121,964.13                | 713.08          | 0.10                      |
| SC     | 842.01                | 98,411.06                 | 703.01          | 0.40                      |
| RAC    | 854.49                | 99,621.09                 | 696.11          | 0.40                      |
| RAE    | 844.79                | 98,847.86                 | 719.06          | 0.12                      |

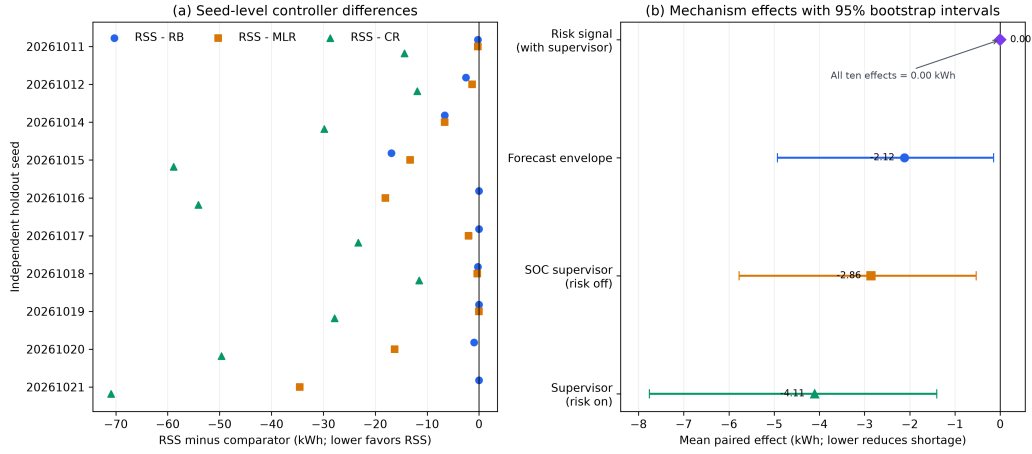


Figure 6: Ten-holdout paired controller differences and mechanism effects

Values displayed as 0.00 are nonnegative recorded times below half of the final reporting unit.

The per-holdout descriptive table below is restricted to RSS, RB, and MLR because it supports the two primary paired contrasts; Table 5 is the complete aggregate comparison of the seven online controllers.

Figure 6(a) summarizes the seed-level controller contrasts for the ten independent holdouts and preserves the seed as the analysis unit. Figure 6(b) reports mean paired mechanism effects with seed-bootstrap 95% intervals. The risk-on minus risk-off effect is exactly 0.00 kWh for every seed, so it is shown explicitly as a zero-width interval rather than as an apparently empty plot.

Table 6: Per-holdout safety and descriptive comparison

| Seed     | RSS<br>unserved<br>(kWh) | RB (kWh) | MLR (kWh) | Cost ratio | Worst regret<br>(kWh) | Audit |
|----------|--------------------------|----------|-----------|------------|-----------------------|-------|
| 20261011 | 104.56                   | 104.81   | 104.78    | 0.99       | 0.00                  | pass  |
| 20261012 | 104.49                   | 107.03   | 105.82    | 0.99       | 0.00                  | pass  |
| 20261014 | 138.81                   | 145.45   | 145.45    | 0.96       | 0.00                  | pass  |
| 20261015 | 159.24                   | 176.15   | 172.50    | 0.93       | 0.00                  | pass  |
| 20261016 | 65.19                    | 65.23    | 83.23     | 0.81       | 0.00                  | pass  |
| 20261017 | 5.48                     | 5.48     | 7.48      | 0.88       | 0.00                  | pass  |
| 20261018 | 71.97                    | 72.18    | 72.31     | 0.97       | 0.00                  | pass  |
| 20261019 | 9.03                     | 9.03     | 9.03      | 0.96       | 0.00                  | pass  |
| 20261020 | 134.97                   | 135.92   | 151.23    | 0.90       | 0.00                  | pass  |
| 20261021 | 19.68                    | 19.68    | 54.22     | 0.49       | 0.00                  | pass  |

Table 6 reports the per-holdout safety and descriptive comparison.

The cost ratio is RSS risk-adjusted cost divided by MLR risk-adjusted cost for the same holdout seed.

### 6.3. Independent causal full-MILP comparator and perfect-foresight reference

The causal rolling full-MILP reference (CR) is implemented independently from the proposed controller. At each 5 s decision point, it solves a 12-step unit-commitment and storage problem and applies the first action. CR uses the same realized plant state, current capacity measurements, CEF point forecast, and cost terms as RSS. It includes integer commitment, minimum stable output, ramping, minimum up/down time, startup and shutdown logic, storage dynamics, and mutually exclusive charging and discharging, but excludes the forecast-error envelope and reserve-SOC supervisor.

CR and RSS share the post-warm-up SOC, generator outputs, committed-unit count, and binding startup and shutdown history. Unit tests verify the minimum up/down constraints at the comparison boundary.

Across ten holdouts, CR produces 1,165.81 kWh of unserved energy, compared with 813.43 kWh for RSS, a reduction of 30.23%. RSS minus CR is  $-35.24 \pm 21.50$  kWh at the seed level (seed-bootstrap 95% interval,  $-48.04$  to  $-22.95$  kWh); all ten differences favor RSS. The exact two-sided sign-flip  $p$ -value is  $< 0.01$  and its Holm-adjusted value is 0.01. Mean CR solve time is 0.01 s per decision and the maximum is 0.19 s. All 30 trajectories have zero recorded power-balance residual and charge-discharge product.

CR also produces more unserved energy than RB, which records 840.96 kWh in Table 5. The controller definitions provide a plausible explanation framework but do not identify the cause of this ordering. CR optimizes a

Table 7: Independent causal and perfect-foresight full-MILP references.

| Reference and scope       | Unserved (kWh) | Solve time / audit                                                  |
|---------------------------|----------------|---------------------------------------------------------------------|
| RSS, ten holdouts         | 813.43         | causal scenario/supervisory pipeline; max 0.36 s/decision; pass     |
| CR, ten holdouts          | 1,165.81       | causal 12-step point forecast; mean 0.01, max 0.19 s/decision; pass |
| PF, original two holdouts | 201.83         | realized 30-min load; mean 0.26 s/window; pass                      |

12-step causal point forecast and has neither the forecast envelope nor the reserve-SOC supervisor; it therefore lacks both explicit forecast-dispersion coverage and the SOC-triggered reserve preparation used by RSS. These differences are sufficient to show that CR is not a like-for-like component ablation. They are not sufficient to attribute the CR-RB gap to a particular mechanism: the available RSS-centered interventions do not decompose CR against RB one change at a time. We therefore treat CR as a bundled causal boundary comparator and do not infer that full-MILP control is intrinsically worse than rule-based control or that either omitted component alone explains the observed gap.

The perfect-foresight reference (PF) solves the same 30 min dispatch problem with the realized load trajectory available over the full horizon. It includes the generator, storage, dynamic-capacity, and cost constraints used in the online comparison, retains a terminal SOC reserve of  $5.00 \times 10^{-3}$ , and uses a linear rated-point fuel approximation. PF minimizes the composite operating objective and provides a best-case reference for the original two holdouts.

On the original two holdouts, PF produces 201.83 kWh of total unserved energy versus 209.05 kWh for RSS. The two PF totals are 82.84 and 118.99 kWh, with optimal solver status for all six scenario windows. The maximum power-balance residual is  $2.30 \times 10^{-13}$  kW, the maximum charge-discharge product is zero, and the mean solve time is 0.26 s per scenario window.

Scopes differ and are shown explicitly in Table 7; RSS is included for context.

#### 6.4. *Orthogonal risk-by-supervisor mechanism audit*

The mechanism audit forms a  $2 \times 2$  design on the same ten seeds, scenario windows, plant, forecast envelope, costs, and solver settings. SC has static scenario CVaR without adaptive risk or the supervisor; RAC adds adaptive risk without the supervisor; RSS-R0 activates the reserve-SOC supervisor while neutralizing only the predicted/dispatch risk level and reserve adder; and RSS activates both. The risk-off intervention holds the SOC thresholds, hysteresis, generator-first reserve logic, grid-headroom rule, capacity traces, and shared warm-up fixed. The 30 newly generated RSS-R0 method-scenario trajectories pass the physical audit; the SC, RAC, and RSS arms reuse the corresponding main-comparison trajectories covered by the main audit.

Within the specified SC/RAC-centered intervention, activating the supervisor is associated with a change in mean seed-level unserved energy by  $-2.86$  kWh (RSS-R0 minus SC); with adaptive risk, its change is  $-4.11$  kWh (RSS minus RAC). Neither effect is adverse on any seed; exact two-sided sign-flip  $p$ -values are 0.03 and 0.02. Adaptive risk without the supervisor changes the mean by  $+1.25$  kWh (RAC minus SC), while the clean marginal risk contrast under the supervisor is 0.00 kWh on every seed. The aggregate risk-adjusted-cost difference between risk-on and risk-off supervision is  $-15.23$  yuan, while shortage is identical to numerical precision.

The trace-level overlap audit is consistent with the null shortage effect. Across 10,800 control steps, high risk occurs on 29.19%, the risk channel changes the internal protection mode on 8.78%, and physical actions differ on 1.72%. Under the implemented controller, the constrained optimizer and reserve-SOC rules absorb most risk-level changes before they reach physical actions. These intervention effects are local to the specified controller configurations and do not constitute an additive decomposition of the RSS-baseline differences.

Figure 7 visualizes why the marginal risk effect is negligible; Table 8 reports the corresponding factorial arms.

#### 6.5. *Forecast uncertainty-envelope ablation*

The paired intervention tests the implemented forecast-envelope bundle over all ten holdouts. It retains the risk signal, SOC supervisor, generator-first execution, and physical constraints, while setting the fixed-reserve and forecast-disagreement terms to zero and replacing all scenario members with the same causal point forecast.

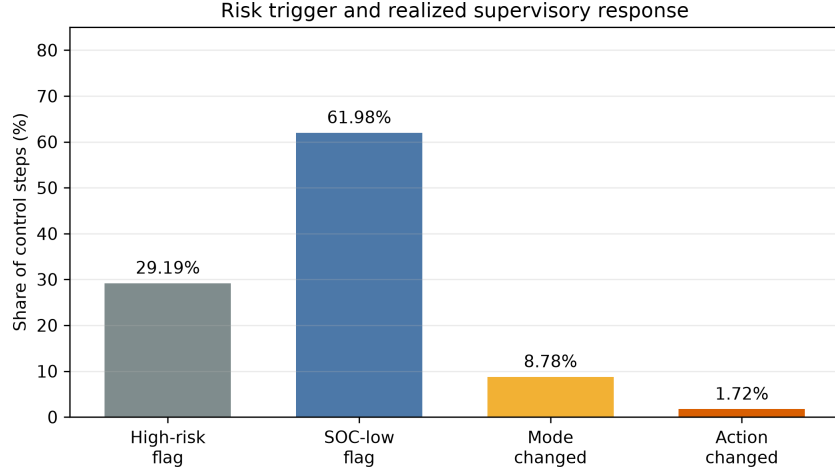


Figure 7: Risk trigger frequency and realized supervisory response

Table 8: Orthogonal risk-by-supervisor mechanism audit over ten holdouts.

| Arm    | Risk | Sup. | Unserved (kWh) | Cost (yuan) | Time (s) |
|--------|------|------|----------------|-------------|----------|
| SC     | off  | off  | 842.01         | 98,411.06   | 0.40     |
| RAC    | on   | off  | 854.49         | 99,621.09   | 0.40     |
| RSS-R0 | off  | on   | 813.43         | 95,382.15   | 0.39     |
| RSS    | on   | on   | 813.43         | 95,366.92   | 0.36     |

Within this paired RSS-centered intervention, the no-envelope variant increases aggregate unserved energy from 813.43 to 834.61 kWh and risk-adjusted cost from 95,366.92 to 97,406.90 yuan, while reducing diesel use from 722.32 to 692.62 L. The paired shortage difference (envelope on minus off) is  $-2.12 \pm 4.34$  kWh per seed, with a seed-bootstrap 95% interval of  $-4.93$  to  $-0.15$  kWh and exact two-sided sign-flip  $p = 0.03$ . Six seeds improve and four tie; none worsens. The registered no-envelope audit covers all 210 method-scenario trajectories and 75,600 rows generated by the full online-controller rerun, whereas the mechanism estimate uses only the paired RSS and RSS-NE outcomes. Within this contrast, enabling the envelope bundle trades additional diesel use for lower shortage and risk-adjusted cost; this local effect does not decompose the full RSS-baseline differences. The no-envelope variant is denoted RSS-NE in Table 9.

Table 9: Ten-holdout uncertainty-envelope intervention.

| Variant | Unserved (kWh) | Risk cost (yuan) | Diesel (L) | Max time (s) | Audit |
|---------|----------------|------------------|------------|--------------|-------|
| RSS     | 813.43         | 95,366.92        | 722.32     | 0.36         | pass  |
| RSS-NE  | 834.61         | 97,406.90        | 692.62     | 0.32         | pass  |

### 6.6. Auxiliary risk-interface diagnostics

The original risk classifier obtains Macro-F1 0.49, high-risk recall 0.78, and severe-risk recall 0.85 on the forecasting replay. Neutralizing this signal in the ten primary holdouts changes aggregate unserved energy by 0.00 kWh, showing no marginal shortage effect for the risk signal in this intervention. This result does not identify the sources of the primary RSS–baseline differences or provide a complete mechanism decomposition.

The auxiliary E-RSS interface is evaluated on a separate six-holdout group. It raises dispatch Macro-F1 from 0.73 to 0.76 and balanced accuracy from 0.75 to 0.79, while increasing the Brier score from 0.05 to 0.11. E-RSS changes 53 of 6,480 physical control rows, but aggregate unserved energy remains 335.23 kWh for both variants and risk-adjusted cost changes from 42,420.31 to 42,419.01 yuan. The interface therefore improves Macro-F1 and balanced accuracy in this group while worsening the Brier score; it exposes source conflict without improving the primary energy-system outcomes.

A separate second six-holdout group evaluates Dempster combination and temperature calibration. DC-RSS reduces mean Brier score from 0.07 for raw D-RSS to 0.03 and raises Macro-F1 from 0.75 for RSS to 0.77. Calibration does not change the selected risk class or dispatch action. Aggregate unserved energy remains 466.05 kWh, and the 4.71 yuan cost difference is negligible relative to the total risk-adjusted cost. Table 10 summarizes the diagnostic variants from this second holdout group. These results support reporting classification and calibration effects separately from dispatch outcomes; they do not provide evidence that the interface caused the primary dispatch improvement reported in Section 6.2.

DC-RSS calibration changes the reported probability vector; its action and control columns are shared with raw D-RSS.

### 6.7. Emergency priority-load ablation

The emergency allocation diagnostic separates protection priority from total supply adequacy. It compares proportional curtailment with a priority-greedy allocator under the same stress scenarios. The two policies produce

Table 10: Auxiliary conflict-rule and probability-calibration diagnostic on the second six-holdout group.

| Variant | Probability layer    | Brier | Macro-F1 | Changed<br>actions vs<br>RSS | Unservd<br>(kWh) |
|---------|----------------------|-------|----------|------------------------------|------------------|
| RSS     | learned output       | 0.03  | 0.75     | 0                            | 466.05           |
| E-RSS   | raw Yager            | 0.10  | 0.77     | 66                           | 466.05           |
| D-RSS   | raw Dempster         | 0.07  | 0.77     | 57                           | 466.05           |
| DC-RSS  | Dempster, $T = 0.50$ | 0.03  | 0.77     | 57                           | 466.05           |

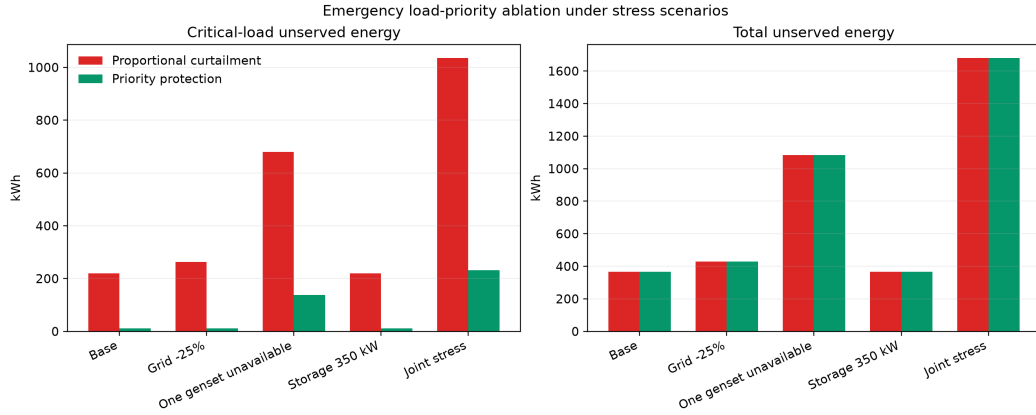


Figure 8: Emergency load-priority ablation

the same total unserved energy in every scenario, while priority protection transfers shortage away from critical loads toward essential and interruptible loads. In the base case, critical-load unserved energy decreases from 219.45 to 11.29 kWh; under the joint extreme stress, it decreases from 1,034.90 to 231.16 kWh. The corresponding total unserved energy remains 366.22 and 1,679.26 kWh, respectively.

Load prioritization changes the distribution of shortage, not the amount of available supply. The class fractions used here are control-study settings that require site-specific electrical and safety approval before deployment.

Figure 8 shows the critical-load and total-shortage comparison across the five stress scenarios; Table 11 reports the corresponding values.

Table 11: Emergency load priority ablation under stress scenarios.

| Scenario                        | Critical<br>unserved:<br>proportional<br>(kWh) | Critical<br>unserved:<br>priority (kWh) | Reduction<br>(kWh) | Total unserved<br>(kWh) |
|---------------------------------|------------------------------------------------|-----------------------------------------|--------------------|-------------------------|
| Base                            | 219.45                                         | 11.29                                   | 208.16             | 366.22                  |
| Grid<br>derating,<br>25%        | 263.57                                         | 11.29                                   | 252.28             | 430.10                  |
| One<br>generator<br>unavailable | 679.87                                         | 137.84                                  | 542.03             | 1,083.29                |
| Storage<br>power cap,<br>350 kW | 219.45                                         | 11.29                                   | 208.16             | 366.22                  |
| Joint<br>extreme<br>stress      | 1,034.90                                       | 231.16                                  | 803.75             | 1,679.26                |

### 6.8. Missing-data and combined-stress diagnostics

The missing-data diagnostic increases MAE from 83.94 kW without injected missingness to 84.33 kW under the moderate condition and 90.66 kW under the severe condition.

Under a combined disturbance involving 25% grid derating, one unavailable generator, a 350 kW storage power cap, a 10 percentage-point SOC bias, a 2.50% load increase, and a 15 s telemetry delay, priority protection reduces critical-load unserved energy from 1,034.90 to 231.16 kWh. Total unserved energy remains 1,679.26 kWh because the allocation layer cannot increase available supply.

Figure 9 shows the detailed joint-extreme trajectory, complementing the cross-scenario comparison in Figure 8.

## 7. Discussion

### 7.1. Energy-system performance

Within the ten simulated holdouts, RSS reduces unserved energy and its shortage-dominated composite cost relative to the prespecified baselines. Relative to RB, it reduces unserved energy by 3.27%, risk-adjusted cost by

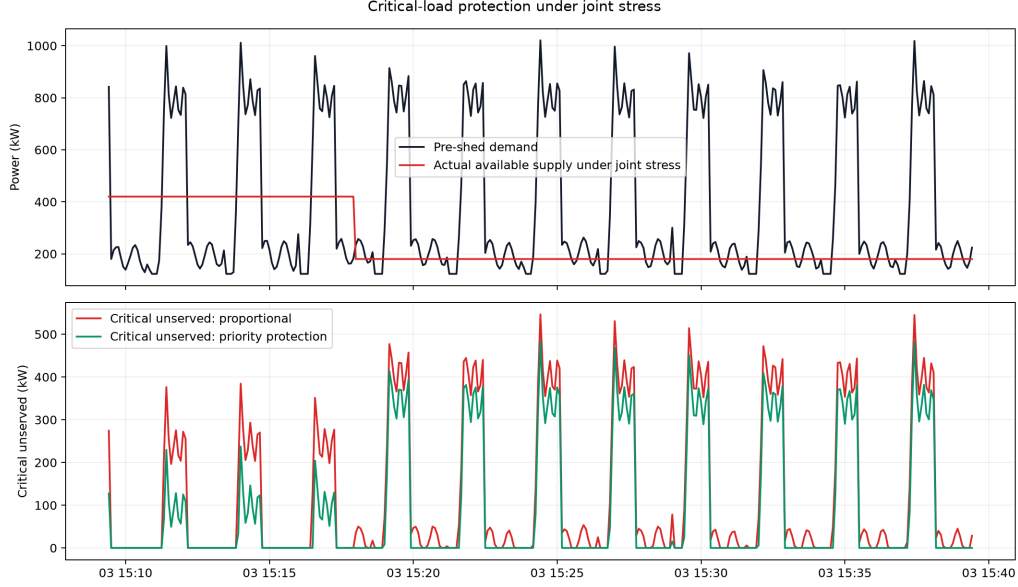


Figure 9: Critical-load protection under joint stress

3.85%, and diesel use by 37.20 L. Relative to MLR, it reduces unserved energy by 10.22% and risk-adjusted cost by 9.34%, while using 2.61 L more diesel. Risk-adjusted cost is not an independent economic outcome. The controller therefore favors supply continuity when forecast transitions and reserve constraints make a small increase in fuel use preferable to additional shortage. All main-comparison trajectories satisfy the power-balance, equipment-capacity, commitment, and SOC checks; intervention audit scopes are reported with their respective analyses.

The forecasting results show that average prediction accuracy alone is an incomplete characterization for this application. iTransformer gives the lowest overall error, while transition-specific errors are reported separately because rapid load changes enter generator-commitment and reserve decisions. The scenario envelope carries this uncertainty into the dispatch problem instead of relying on a single point forecast.

## 7.2. Operating mechanisms

Two prespecified component interventions are associated with lower unserved energy within the simulated protocol. First, enabling the implemented forecast- envelope bundle is associated with a 2.12 kWh mean reduction per

holdout in the paired RSS-centered intervention. Second, activating the SOC supervisor is associated with a 2.86–4.11 kWh mean reduction per holdout in the specified SC/RAC-centered contrasts. The controller rules provide a consistent operational interpretation: they prepare reachable generator capacity before the battery reaches its lower operating region and avoid releasing protection immediately after a short recovery.

The adaptive risk signal has no marginal aggregate-shortage effect in the specified intervention. In the auxiliary studies, the raw evidential variant improves selected classification metrics while worsening the Brier score, and temperature calibration improves the Brier score in a separate holdout group; neither changes unserved energy. These results narrow the role of the risk interface to separately reported operator-facing information. The forecast-envelope bundle and reserve–SOC logic are associated with reductions in their corresponding interventions. Likewise, priority-based allocation protects critical loads during severe disturbances but cannot reduce total shortage when aggregate supply is fixed. These results are not an additive decomposition of the RSS–RB, RSS–MLR, or RSS–CR differences.

### *7.3. Engineering applicability*

Earlier drilling rig studies established the value of rule-based generator scheduling and battery storage. The present framework adds forecast uncertainty, discrete multi-generator constraints, explicit storage reserve, and priority-based protection in a single rolling controller. The resulting structure may offer a transferable control architecture for other industrial grid–diesel–battery systems with abrupt loads and limited grid capacity, but such transfer requires system-specific model identification, parameterization, safety approval, and validation.

Deployment requires synchronized total active power, individual generator states and outputs, battery SOC and power, grid capacity measurements, fuel consumption, energy prices, and approved load classes. Equipment ramp limits, minimum up/down times, efficiency curves, startup costs, degradation costs, and terminal SOC requirements should be replaced with site values. The next validation step is an external replay with site data followed by hardware-in-the-loop testing on the intended control platform.

## 8. Limitations and reproducibility

The evaluation record is generated by one mechanistic simulation parameterized from public evidence; synchronized target-rig electrical measurements were not available. The ten holdouts test controller behavior across independently generated cases but do not capture installation-specific equipment dynamics, communication delays, or operator actions. The forecasting comparison also uses a fixed retrospective replay rather than an external prospective dataset. The holdouts are pseudorandom realizations of the same frozen simulator, not samples from an empirically fitted target-rig population. The seven forecast members likewise are deterministic model-disagreement trajectories rather than probability-calibrated samples; with their equal weights, the reported CVaR levels select the same worst member and do not establish probabilistic coverage.

The bundled causal full-MILP boundary comparator omits the forecast envelope and supervisory policy, while the perfect-foresight solution is available only for the original two cases. These references compare different declared information and control bundles and do not support attribution to any single component; they also do not exhaust all possible robust or stochastic optimization baselines. The emergency load classes and equipment parameters also require site-specific approval and calibration.

The source artifacts, configurations, result manifests, and analysis scripts are versioned locally. A public release of the permitted data generator and reproducibility package should accompany the final article or be deposited in a suitable repository.

## 9. Conclusions

This study evaluated reserve-SOC supervisory energy management using a 72 h record and ten independently generated holdouts from a single mechanistic drilling-load and power-system simulator; no synchronized measurements from the target rig were used. Seven online controllers shared the simulated plant, realized traces, price terms, execution layer, and audit path; their forecast representations and terminal-SOC rules followed their declared controller-specific definitions.

Across ten holdouts, RSS reduces unserved energy by 3.27% relative to rule-based control, 10.22% relative to deterministic margin-loaded reference MPC (MLR), and 30.23% relative to the causal full-MILP boundary comparator. Risk-adjusted cost falls by 3.85% relative to rule-based control and

9.34% relative to MLR; these shortage-dominated composite cost comparisons are not independent economic evidence. The causal full-MILP is independently implemented, uses a single point forecast, and omits both the forecast-error envelope and reserve-SOC supervisor. Because it is weaker than rule-based control under this protocol, the 30.23% RSS-CR gap is a bundled boundary comparison rather than being attributable to any single component or a like-for-like ablation. Component-level interventions associate the forecast-envelope bundle and SOC supervision with local reductions in their corresponding interventions, while the adaptive risk signal in the specified intervention has no marginal shortage effect. All main-comparison trajectories satisfy the stated physical constraints, and the maximum recorded decision time is 0.36 s.

Within this simulated system, the results support coordinating storage reserve with forecast uncertainty and reachable generator capacity during rapid load transitions. They do not establish performance for the target rig or for drilling rigs generally. Site-specific electrical data and hardware-in-the-loop testing remain necessary before field-performance conclusions can be drawn.

#### **CRediT authorship contribution statement**

**Shixing Zhang:** Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Writing–original draft, Project administration.

**Pengchong Wei:** Methodology, Validation, Investigation, Data curation, Writing–review & editing.

**Lei Luo:** Resources, Validation, Writing–review & editing.

**Pengfei Zhang:** Investigation, Validation, Writing–review & editing.

**Xingjun Yu:** Supervision, Project administration, Funding acquisition, Resources, Writing–review & editing.

#### **Declaration of competing interest**

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interpretation, manuscript preparation, or the decision to submit the article. The authors declare no other competing financial interests or personal relationships that could have influenced the work reported in this paper.

## Data availability

The compact reproducibility package is available in the public GitHub repository at <https://github.com/shixingzhang1994-byte/rig-energy-forecasting/tree/reproducibility-package-v1>. It contains the permitted mechanistic simulator and data generator, controller and experiment configurations, environment specification, the simulated 72 h record, selected trajectory-level audit outputs, result manifests, analysis scripts, and manuscript sources supporting the reported comparisons. Large development intermediates, duplicate plots, and model weights are excluded as documented in the repository README. A DOI-bearing archival release and its reuse license will be added after author and institutional confirmation.

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## References

- [1] H. Jokar, B. Bahmani-Firouzi, M. Simab, Bilevel model for security-constrained and reliability transmission and distribution substation energy management considering large-scale energy storage and demand side management, *Energy Reports* 8 (2022) 2617–2629. doi:10.1016/j.egyr.2022.01.137.  
URL <https://doi.org/10.1016/j.egyr.2022.01.137>
- [2] N. Vafamand, J. Boudjadar, M. H. Khooban, Model predictive energy management in hybrid ferry grids, *Energy Reports* 6 (2020) 550–557. doi:10.1016/j.egyr.2019.11.118.  
URL <https://doi.org/10.1016/j.egyr.2019.11.118>

- [3] S. Vergine, C. Álvarez-Arroyo, G. D’Amico, J. M. Escaño, L. Alvarado-Barrios, Optimal management of a hybrid and isolated microgrid in a random setting, *Energy Reports* 8 (2022) 9402–9419. doi:10.1016/j.egyr.2022.07.044.  
URL <https://doi.org/10.1016/j.egyr.2022.07.044>
- [4] D. Pavković, A. Sedić, Z. Guzović, Oil drilling rig diesel power-plant fuel efficiency improvement potentials through rule-based generator scheduling and utilization of battery energy storage system, *Energy Conversion and Management* 121 (2016) 194–211. doi:10.1016/j.enconman.2016.05.022.  
URL <https://doi.org/10.1016/j.enconman.2016.05.022>
- [5] E. Chupin, K. Frolov, M. Korzhavin, O. Zhdaneev, Energy storage systems for drilling rigs, *Journal of Petroleum Exploration and Production Technology* 12 (2) (2021) 341–350. doi:10.1007/s13202-021-01248-5.  
URL <https://doi.org/10.1007/s13202-021-01248-5>
- [6] Y. Huang, H. Lan, Y.-Y. Hong, S. Wen, H. Yin, Optimal generation scheduling for a deep-water semi-submersible drilling platform with uncertain renewable power generation and loads, *Energy* 181 (2019) 897–907. doi:10.1016/j.energy.2019.05.157.  
URL <https://doi.org/10.1016/j.energy.2019.05.157>
- [7] C. S. Peixoto, G. G. T. T. Vieira, M. B. C. Salles, B. S. Carmo, Assessing the impact of power dispatch optimization and energy storage systems in diesel-electric PSVs: A case study based on real field data, *Applied Energy* 357 (2024) 122476. doi:10.1016/j.apenergy.2023.122476.  
URL <https://doi.org/10.1016/j.apenergy.2023.122476>
- [8] R. Banihabib, M. Assadi, Towards a low-carbon future for offshore oil and gas industry: A smart integrated energy management system with floating wind turbines and gas turbines, *Journal of Cleaner Production* 423 (2023) 138742. doi:10.1016/j.jclepro.2023.138742.  
URL <https://doi.org/10.1016/j.jclepro.2023.138742>
- [9] R. Zamora, A. K. Srivastava, Controls for microgrids with storage: Review, challenges, and research needs, *Renewable and Sustainable Energy Reviews* 14 (7) (2010) 2009–2018. doi:10.1016/j.rser.2010.03.019.  
URL <https://doi.org/10.1016/j.rser.2010.03.019>

- [10] L. Meng, E. R. Sanseverino, A. Luna, T. Dragicevic, J. C. Vasquez, J. M. Guerrero, Microgrid supervisory controllers and energy management systems: A literature review, *Renewable and Sustainable Energy Reviews* 60 (2016) 1263–1273. doi:10.1016/j.rser.2016.03.003. URL <https://doi.org/10.1016/j.rser.2016.03.003>
- [11] M. F. Zia, E. Elbouchikhi, M. Benbouzid, Microgrids energy management systems: A critical review on methods, solutions, and prospects, *Applied Energy* 222 (2018) 1033–1055. doi:10.1016/j.apenergy.2018.04.103. URL <https://doi.org/10.1016/j.apenergy.2018.04.103>
- [12] A. R. Abbasi, D. Baleanu, Recent developments of energy management strategies in microgrids: An updated and comprehensive review and classification, *Energy Conversion and Management* 297 (2023) 117723. doi:10.1016/j.enconman.2023.117723. URL <https://doi.org/10.1016/j.enconman.2023.117723>
- [13] O. N. Onsomu, E. Terciyanlı, B. Yeşilata, Comprehensive review of energy management strategies: Considering battery energy storage system and renewable energy sources, *Engineering Reports* 6 (11) (2024). doi:10.1002/eng2.12995. URL <https://doi.org/10.1002/eng2.12995>
- [14] T. Pamulapati, M. Cavus, I. Odigwe, A. Allahham, S. Walker, D. Giouris, A review of microgrid energy management strategies from the energy trilemma perspective, *Energies* 16 (1) (2023) 289. doi:10.3390/en16010289. URL <https://doi.org/10.3390/en16010289>
- [15] K. S. Joshal, N. Gupta, Microgrids with model predictive control: A critical review, *Energies* 16 (13) (2023) 4851. doi:10.3390/en16134851. URL <https://doi.org/10.3390/en16134851>
- [16] J. Hu, Y. Shan, Y. Yang, A. Parisio, Y. Li, N. Amjady, S. Islam, K. W. Cheng, J. M. Guerrero, J. Rodríguez, Economic model predictive control for microgrid optimization: A review, *IEEE Transactions on Smart Grid* 15 (1) (2024) 472–484. doi:10.1109/TSG.2023.3266253. URL <https://doi.org/10.1109/TSG.2023.3266253>

- [17] F. J. Vivas, A. Pajares, X. Blasco, J. M. Herrero, F. Segura, J. M. Andújar, A novel energy management system based on two-level hierarchical economic model predictive control for use in microgrid control, *Energy Conversion and Management*: X 26 (2025) 101027. doi:10.1016/j.ecmx.2025.101027.  
URL <https://doi.org/10.1016/j.ecmx.2025.101027>
- [18] D. Lim, J. Cha, C. Seo, G. Jang, A two-layer robust model predictive control-based energy management system for a zero-carbon DC microgrid with hybrid hydrogen-battery energy storage, *International Journal of Electrical Power & Energy Systems* 174 (2026) 111516. doi:10.1016/j.ijepes.2025.111516.  
URL <https://doi.org/10.1016/j.ijepes.2025.111516>
- [19] J. Hu, Y. Shan, J. M. Guerrero, A. Ioinovici, K. W. Chan, J. Rodriguez, Model predictive control of microgrids – an overview, *Renewable and Sustainable Energy Reviews* 136 (2021) 110422. doi:10.1016/j.rser.2020.110422.  
URL <https://doi.org/10.1016/j.rser.2020.110422>
- [20] Y. Zhang, T. Zhang, R. Wang, Y. Liu, B. Guo, Optimal operation of a smart residential microgrid based on model predictive control by considering uncertainties and storage impacts, *Solar Energy* 122 (2015) 1052–1065. doi:10.1016/j.solener.2015.10.027.  
URL <https://doi.org/10.1016/j.solener.2015.10.027>
- [21] L. A. Dao, A. Dehghani-Pilehvarani, A. Markou, L. Ferrarini, A hierarchical distributed predictive control approach for microgrids energy management, *Sustainable Cities and Society* 48 (2019) 101536. doi:10.1016/j.scs.2019.101536.  
URL <https://doi.org/10.1016/j.scs.2019.101536>
- [22] F. Garcia-Torres, C. Bordons, M. A. Ridao, Optimal economic schedule for a network of microgrids with hybrid energy storage system using distributed model predictive control, *IEEE Transactions on Industrial Electronics* 66 (3) (2019) 1919–1929. doi:10.1109/TIE.2018.2826476.  
URL <https://doi.org/10.1109/TIE.2018.2826476>
- [23] J. Tobajas, F. Garcia-Torres, P. Roncero-Sanchez, J. Vazquez, L. Bellatreche, E. Nieto, Resilience-oriented schedule of microgrids with hybrid

- energy storage system using model predictive control, *Applied Energy* 306 (2022) 118092. doi:10.1016/j.apenergy.2021.118092.  
URL <https://doi.org/10.1016/j.apenergy.2021.118092>
- [24] G. Franke, M. Schneider, T. Weitzel, S. Rinderknecht, Stochastic optimization model for energy management of a hybrid microgrid using mixed integer linear programming, *IFAC-PapersOnLine* 53 (2) (2020) 12948–12955. doi:10.1016/j.ifacol.2020.12.2132.  
URL <https://doi.org/10.1016/j.ifacol.2020.12.2132>
- [25] L. Ji, G. Huang, Y. Xie, Y. Zhou, J. Zhou, Robust cost-risk tradeoff for day-ahead schedule optimization in residential microgrid system under worst-case conditional value-at-risk consideration, *Energy* 153 (2018) 324–337. doi:10.1016/j.energy.2018.04.037.  
URL <https://doi.org/10.1016/j.energy.2018.04.037>
- [26] R. T. Rockafellar, S. Uryasev, Optimization of conditional value-at-risk, *The Journal of Risk* 2 (3) (2000) 21–41. doi:10.21314/JOR.2000.038.  
URL <https://doi.org/10.21314/JOR.2000.038>
- [27] L. O. P. Vásquez, J. L. Redondo, J. D. Á. Hervás, V. M. Ramírez, J. L. Torres, Balancing CO2 emissions and economic cost in a microgrid through an energy management system using MPC and multi-objective optimization, *Applied Energy* 347 (2023) 120998. doi:10.1016/j.apenergy.2023.120998.  
URL <https://doi.org/10.1016/j.apenergy.2023.120998>
- [28] A. Pajares, F. J. Vivas, X. Blasco, J. M. Herrero, F. Segura, J. M. Andújar, Methodology for energy management strategies design based on predictive control techniques for smart grids, *Applied Energy* 351 (2023) 121809. doi:10.1016/j.apenergy.2023.121809.  
URL <https://doi.org/10.1016/j.apenergy.2023.121809>
- [29] R. Seppehrzad, J. Ghafourian, A. Hedayatnia, A. Al-Durrad, M. H. Khooban, Experimental and developed DC microgrid energy management integrated with battery energy storage based on multiple dynamic matrix model predictive control, *Journal of Energy Storage* 74 (2023) 109282. doi:10.1016/j.est.2023.109282.  
URL <https://doi.org/10.1016/j.est.2023.109282>

- [30] M. A. Alarcón, R. G. Alarcón, A. H. González, A. Ferramosca, A scenario-based economic-stochastic model predictive control for the management of microgrids, *Sustainable Energy, Grids and Networks* 36 (2023) 101205. doi:10.1016/j.segan.2023.101205.  
URL <https://doi.org/10.1016/j.segan.2023.101205>
- [31] T. Ábelová, R. Kohút, K. Fedorová, M. Kvasnica, Risk-aware stochastic energy management of microgrid with battery storage and renewables, *IFAC-PapersOnLine* 56 (2) (2023) 8445–8450. doi:10.1016/j.ifacol.2023.10.1042.  
URL <https://doi.org/10.1016/j.ifacol.2023.10.1042>
- [32] C. Zhang, H. Liang, L. Yang, A distributionally robust approximate framework consider CVaR constraints for energy management of microgrid, *Sustainable Energy, Grids and Networks* 36 (2023) 101172. doi:10.1016/j.segan.2023.101172.  
URL <https://doi.org/10.1016/j.segan.2023.101172>
- [33] R. R. Yager, On the Dempster–Shafer framework and new combination rules, *Information Sciences* 41 (2) (1987) 93–137. doi:10.1016/0020-0255(87)90007-7.  
URL [https://doi.org/10.1016/0020-0255\(87\)90007-7](https://doi.org/10.1016/0020-0255(87)90007-7)
- [34] T. Denœux, Decision-making with belief functions: A review, *International Journal of Approximate Reasoning* 109 (2019) 87–110. doi:10.1016/j.ijar.2019.03.009.  
URL <https://doi.org/10.1016/j.ijar.2019.03.009>
- [35] M. Demirci, M. Saka, H. Gozde, M. Dursun, M. C. Taplamacioglu, Dempster–Shafer evidence theory application for fault diagnosis of power transformers, in: 2022 9th International Conference on Electrical and Electronics Engineering (ICEEE), IEEE, 2022, pp. 40–44. doi:10.1109/ICEEE55327.2022.9772608.  
URL <https://doi.org/10.1109/ICEEE55327.2022.9772608>
- [36] X. Wu, H. Zhao, W. Xu, W. Pan, Q. Ji, X. Hua, Fault diagnosis of the distribution network based on the D–S evidence theory Bayesian network, *Frontiers in Energy Research* 12 (2024) 1422639. doi:10.3389/fenrg.2024.1422639.  
URL <https://doi.org/10.3389/fenrg.2024.1422639>

- [37] X. Guan, H. Yu, X. Yi, Optimized methods for basic probability assignments in evidence theory: Applications to fault diagnosis, *Engineering Applications of Artificial Intelligence* 160 (2025) 111737. doi:10.1016/j.engappai.2025.111737.  
URL <https://doi.org/10.1016/j.engappai.2025.111737>
- [38] M. Esmaili, A. A. Ahmadi, A. Nateghi, M. Shafie-khah, Robust power management system with generation and demand prediction and critical loads in DC microgrid, *Journal of Cleaner Production* 384 (2023) 135490. doi:10.1016/j.jclepro.2022.135490.  
URL <https://doi.org/10.1016/j.jclepro.2022.135490>
- [39] A. K. Erenoglu, O. Erding, Real-time allocation of multi-mobile resources in integrated distribution and transportation systems for resilient electrical grid, *IEEE Transactions on Power Delivery* 38 (2) (2023) 1108–1119. doi:10.1109/TPWRD.2022.3206272.  
URL <https://doi.org/10.1109/TPWRD.2022.3206272>
- [40] A. Niknami, M. Tolou Askari, M. Amir Ahmadi, M. Babaei Nik, M. Samiei Moghaddam, Resilient day-ahead microgrid energy management with uncertain demand, EVs, storage, and renewables, *Cleaner Engineering and Technology* 20 (2024) 100763. doi:10.1016/j.clet.2024.100763.  
URL <https://doi.org/10.1016/j.clet.2024.100763>
- [41] G. Maślak, P. Orłowski, Operational optimisation of a microgrid using non-stationary hybrid switched model predictive control with virtual storage-based demand management, *Renewable and Sustainable Energy Reviews* 202 (2024) 114685. doi:10.1016/j.rser.2024.114685.  
URL <https://doi.org/10.1016/j.rser.2024.114685>
- [42] B. Lim, S. Ö. Arık, N. Loeff, T. Pfister, Temporal fusion transformers for interpretable multi-horizon time series forecasting, *International Journal of Forecasting* 37 (4) (2021) 1748–1764. doi:10.1016/j.ijforecast.2021.03.012.  
URL <https://doi.org/10.1016/j.ijforecast.2021.03.012>
- [43] Y. Nie, N. H. Nguyen, P. Sinthong, J. Kalagnanam, A time series is worth 64 words: Long-term forecasting with transformers (2023).

- arXiv:2211.14730.  
URL <https://arxiv.org/abs/2211.14730>
- [44] Y. Liu, T. Hu, H. Zhang, H. Wu, S. Wang, L. Ma, M. Long, itransformer: Inverted transformers are effective for time series forecasting (2024). arXiv:2310.06625.  
URL <https://arxiv.org/abs/2310.06625>
  - [45] Y. Eren, İ. Küçükdemiral, A comprehensive review on deep learning approaches for short-term load forecasting, *Renewable and Sustainable Energy Reviews* 189 (2024) 114031. doi:10.1016/j.rser.2023.114031.  
URL <https://doi.org/10.1016/j.rser.2023.114031>
  - [46] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, W. Zhang, Informer: Beyond efficient transformer for long sequence time-series forecasting, *Proceedings of the AAAI Conference on Artificial Intelligence* 35 (12) (2021) 11106–11115. doi:10.1609/AAAI.V35I12.17325.  
URL <https://doi.org/10.1609/AAAI.V35I12.17325>
  - [47] H. Wu, J. Xu, J. Wang, M. Long, Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting, in: M. Ranzato, A. Beygelzimer, Y. Dauphin, P. S. Liang, J. Wortman Vaughan (Eds.), *Advances in Neural Information Processing Systems*, Vol. 34, Curran Associates, Inc., 2021, pp. 22419–22430.  
URL [https://proceedings.neurips.cc/paper\\_files/paper/2021/hash/bcc0d400288793e8bdcd7c19a8ac0c2b-Abstract.html](https://proceedings.neurips.cc/paper_files/paper/2021/hash/bcc0d400288793e8bdcd7c19a8ac0c2b-Abstract.html)
  - [48] T. Zhou, Z. Ma, Q. Wen, X. Wang, L. Sun, R. Jin, FEDformer: Frequency enhanced decomposed transformer for long-term series forecasting, in: K. Chaudhuri, S. Jegelka, L. Song, C. Szepesvari, G. Niu, S. Sabato (Eds.), *Proceedings of the 39th International Conference on Machine Learning*, Vol. 162 of *Proceedings of Machine Learning Research*, PMLR, 2022, pp. 27268–27286.  
URL <https://proceedings.mlr.press/v162/zhou22g.html>
  - [49] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, M. Long, Timesnet: Temporal 2d-variation modeling for general time series analysis, Published at the Eleventh International Conference on Learning Representations (2023). arXiv:2210.02186.  
URL <https://arxiv.org/abs/2210.02186>